

Exercise

Part 1: Regulations and Data Science Ethics [15 Marks]

By conducting your own research, identify the key principles that must be followed to guarantee the ethical use of algorithms, with particular attention to issues of privacy, fairness, and solidarity. In addition, provide a concise summary of the strategies and practices that insurance companies can adopt to protect individuals' privacy, maintain fairness in algorithmic decision-making, and foster solidarity within algorithmic frameworks. Be sure to cite all relevant academic articles and sources used to support your discussion.

[15 marks]

Part 2: Data Manipulation and Wrangling [25 Marks]

For this question, you are given a **travel insurance** data set called `travel_insurance` with the following columns:

- **caseId**: a numeric column used for identification.
- **attorney**: whether the claimant is represented by an attorney (1 for no and 2 for yes).
- **claimantGender**: the claimant's gender (1 for male and 2 for female).
- **incident_type**: the type of travel incident (0 for minor, 1 for severe, 2 if not applicable).
- **claimantAge**: the age of the claimant.
- **loss**: the total financial loss incurred to the claimant in thousands.

- (a) Display the structure of the data set `travel_insurance` and compute summary statistics for each of its columns.

[1 mark]

- (b) Calculate the range, interquartile range, 0.5 and 99.5 percentile of the `loss` column. Are there any obvious outliers? Replace these entries with NA. [3 marks]
- (c) Delete the entries related to outlying `loss` values above. In insurance, transactional data might appear in a data set as a negative claim amount. Whilst there are uses for this, we do not need this right now. Remove the rows with negative values in `loss`. [7 marks]
- (d) Perform a Box-Cox transformation on `loss`, and record this on a separate column. [4 marks]
- (e) Compute the mean claim values amongst men and women. [4 marks]
- (f) Create a categorical variable for the age of the claimants, with the following groups: (strictly) under 25, from 26 to 35, from 36–42, from 43–72, and 72 or above. [6 marks]

Part 3: Principal Component Analysis (PCA) and Linear Regression [30 Marks]

You are given a data set called `property`. It contains the following variables:

- **propertyVal**: insured property value (in thousands).
- **claimCount**: number of property claims.
- **claimCost**: total claim cost (in thousands).
- **propertyAge**: age of the property (in years).
- **distance**: average distance to the nearest emergency service (in kilometres).
Note: Including this variable reflects how insurers use risk zoning: premiums are higher for properties with limited access to emergency response, especially in rural or remote areas.
- **duration**: duration of the insurance policy (in years).

- (a) Identify which two variables are most correlated. [3 marks]

- (b) Perform a PCA using the `princomp()` function in R and determine the minimum number of principal components required to retain 80% of the total variation in the data. [10 marks]
- (c) Produce a PCA plot. [3 marks]
- (d) Fit a linear regression model using the principal components selected above. You do not need to perform model selection or residual analysis. [6 marks]
- (e) A new policy is issued with the following details:

variable	value
propertyVal	22
propertyAge	14.5
distance	25
duration	10

Using the regression model fitted above, predict the `claimCost` for this policy. [8 marks]

Part 4: Deep Neural Networks [30 Marks]

Load the data called `freMTPL2freq` which is given to you. Response variable in this data set is the number of claims and contains several explanatory variables.

- (a) Display the structure of the data set and use a Min-Max scaler to normalize all continuous predictors between -1 and 1 and embed the categorical feature components into embedding layers. [3 marks]
- (b) Construct a 3-layer network with neurons $(q1, q2, q3) = (32, 24, 16)$ and employ the hyperbolic tangent activation function. Additionally, incorporate a non-trainable offset, $\log(\text{Exposure})$. Proceed by utilizing the Poisson deviance loss as the objective function, and outline the Nadam optimizer. Subsequently, initiate the training of the network.

[15 marks]

- (c) Explain the training process for a network with a structure resembling the one mentioned earlier, utilizing the deviance loss of the Poisson–Inverse Gaussian distribution with parameters $\mu > 0$ as the mean and $\sigma > 0$, and probability mass function

$$\mathbb{P}(k \mid \mu, \sigma) = \left(\frac{2\alpha}{\pi}\right)^{1/2} \frac{e^{1/\sigma}}{k!} \left(\frac{\mu}{\alpha\sigma}\right)^k K_{k-\frac{1}{2}}(\alpha), \quad (1)$$

where $\alpha^2 = \frac{1}{\sigma^2} + \frac{2\mu}{\sigma}$, for $k = 0, 1, 2, \dots$ denoting the number of claims.

The function $K_\lambda(t)$ is the modified Bessel function of the third kind, defined as

$$K_\lambda(t) = \frac{1}{2} \int_0^\infty x^{\lambda-1} \exp\left\{-\frac{1}{2}t\left(x + \frac{1}{x}\right)\right\} dx. \quad (2)$$

For this part, assume $\sigma = 1.2$ throughout.

Provide all the relevant mathematical expressions for training the network. Note in question (c) you are not asked to implement any algorithm in language R.

[12 marks]